

EXAM

Video Processing
Course number 0512-4263-01
Spring 2022
Faculty of Engineering
Tel-Aviv University
Prof. Shai Avidan

DATE: 19/6/2022
TIME: 2pm
DURATION: 3 hours

Read all of the following information before starting the exam:

- Show all work, clearly and in order.
- Justify your answers.
- Please keep your written answers brief; be clear and to the point.
- Each answer should take one page only (text, images and equations).
- This exam has 8 questions and is worth 100 points.
- Good luck!



Figure 1: (Left) Input image. (Right) Semantic Segmentation result.

1. (13 points) OPTIC FLOW

Assume you have access to a semantic segmentation neural network that given an input image I computes its corresponding semantic segmentation map S . See example in Figure 1. Each pixel in S holds a value in the range $\{0, \dots, c - 1\}$ where c is the number of classes (i.e., car, road, pedestrian, etc...). Observe that all instances of the same class receive the same label. For example, all vehicles are labeled with the same label.

Modify the Horn-Schunck algorithm to take as input two grayscale images I_1 and I_2 and their corresponding semantic segmentation maps S_1 and S_2 , and produce the optic flow between them.



2. (12 points) TRACKING

We wish to simulate rolling a loaded (unfair) six-sided (cube-shape) die, whose probabilities are $(p_1, \dots, p_6)$. We only have access to the function *rand()* that returns a pseudo-random number r sampled uniformly in the range $[0, 1]$. Suggest a way to run the simulation using *rand()*.

3. (13 points) INTRINSIC IMAGES

Suggest an *online* algorithm for computing the intrinsic reflectance image R_t from a sequence of images $I_1, \dots, I_t$. The algorithm can store as many as 256 input images (or their equivalent in terms of memory footprint). Hint: the intensity values of a pixel in images $I_1, \dots, I_t$ are in the range $[0, 255]$.





Figure 2: A frame from a video of the Iguazu falls. In the video, the camera is panning from left to right. Suggest a way to synthesize a video in which the camera is panning right to left and still have the water falling down.

4. (12 points) EVOLVING TIME FRONTS

You are given a video sequence of a camera panning left-to-right that captures the Iguazu waterfalls. See a sample frame in Figure 2. Explain how to create a video sequence of the camera panning right-to-left with the water flowing down. (This problem appeared in the paper discussed in class.)

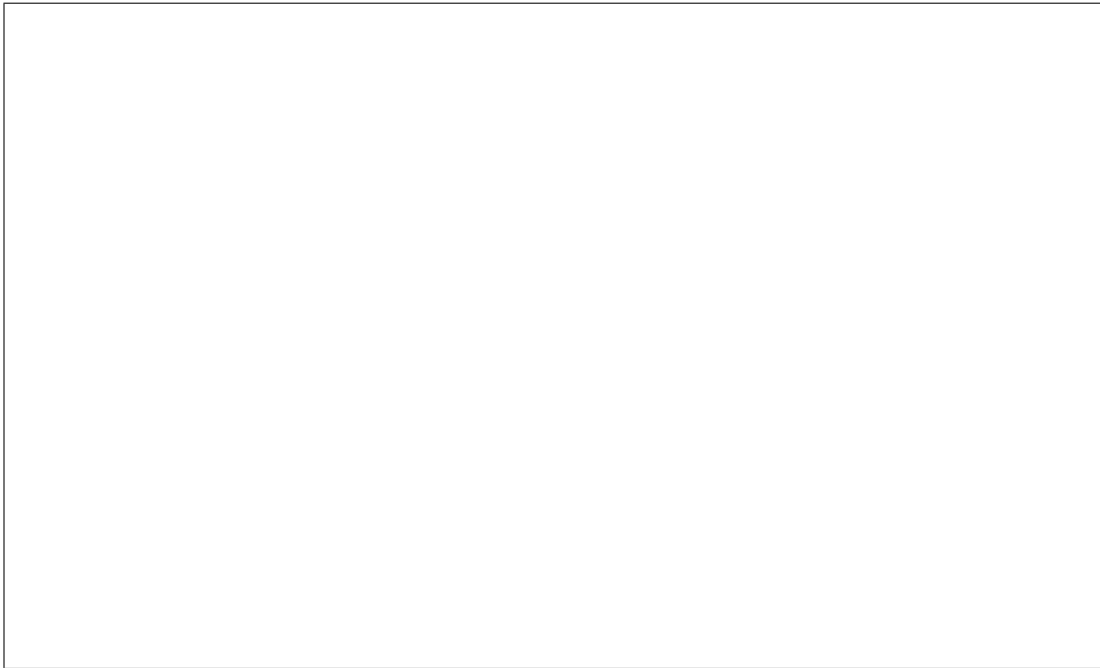


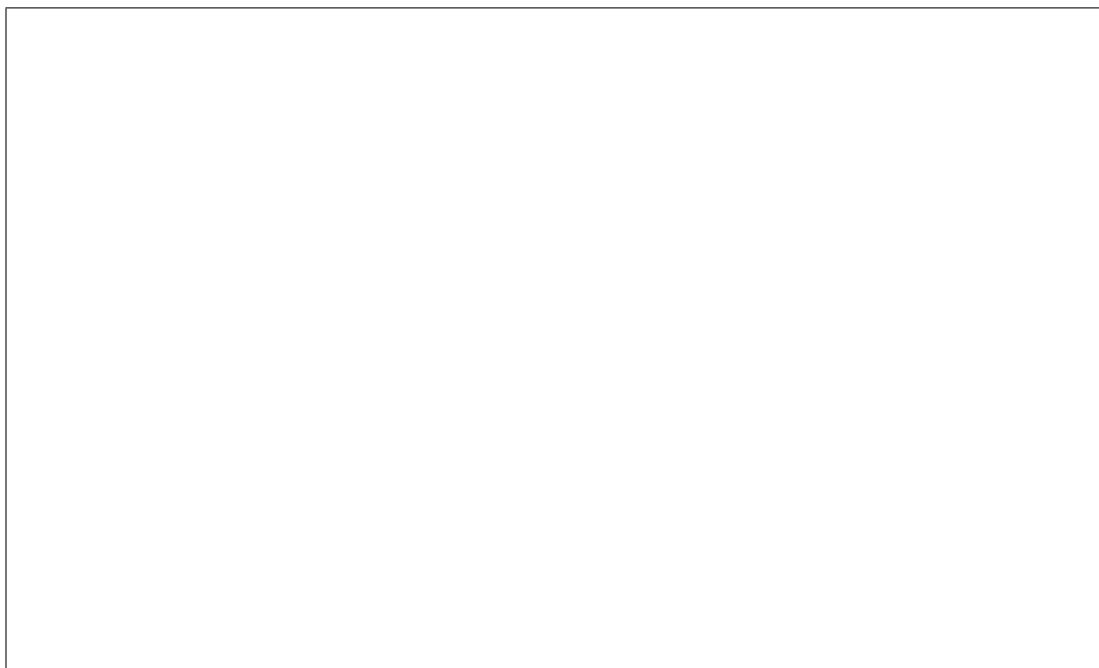


Figure 3: (Left) Original Image. (Center) Input ("Doctored") image. (Right) Result of clone detection algorithm. The algorithm takes as input the *center* image and automatically detects cloning (i.e., the two dark regions are clones). The original image is shown only for reference and is not used by the algorithm.

5. (13 points) CLONE DETECTION

Consider Figure 3. The user took the original image (on the left) and used inpainting to remove the garbage can, resulting in the "Doctored" image in the middle. Suggest a clone detection algorithm that takes an image as input (i.e., the *middle* image of the panel) and automatically detects cloned regions (i.e., the mask on the right that shows that the two dark regions are clones). The original image is shown on the left for reference only. The algorithm has *no access* to it.

Suggest an algorithm for clone detection.



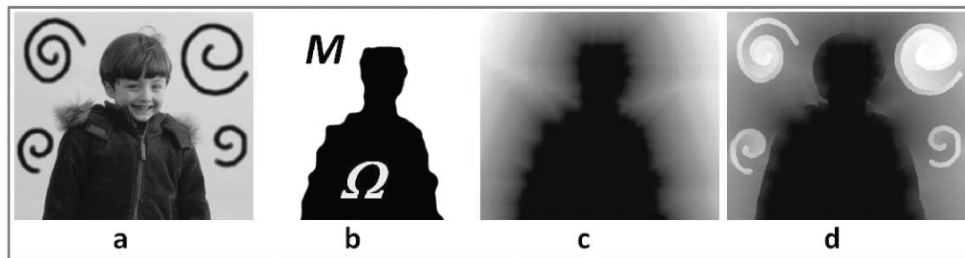


Figure 4: (a) Input image I . (b) Input mask M with "object" region Ω . (c,d) Two distance images $D_l(\mathbf{x}; M, \nabla I)$ with two different γ values. Determine for (c) and (d) if they are using $\gamma = 0$ or $\gamma > 0$.

6. (12 points) GEODESIC DISTANCE

Let $I(\mathbf{x}) : \Psi \rightarrow \mathbb{R}^d$ be an image (with $d = 3$ for a color image), with support $\Psi \subset \mathbb{R}^2$. Given a binary mask M (with $M(\mathbf{x}) \in \{0, 1\}, \forall \mathbf{x} \in \Psi$) associated to a "seed" region (or "object" region) $\Omega = \{\mathbf{x} \in \Psi : M(\mathbf{x}) = 0\}$, the unsigned Geodesic Distance $D_l(\mathbf{x}; M, \nabla I)$ is given by

$$D_l(\mathbf{x}; M, \nabla I) = \min_{\{\mathbf{x}' | M(\mathbf{x}') = 0\}} d(\mathbf{x}, \mathbf{x}'), \text{ with} \quad (1)$$

$$d(\mathbf{a}, \mathbf{b}) = \inf_{\Gamma \in \mathcal{P}_{\mathbf{a}, \mathbf{b}}} \int_0^{l(\Gamma)} \sqrt{1 + \gamma^2 (\nabla I(s) \cdot \Gamma'(s))^2} ds, \quad (2)$$

where $\mathcal{P}_{\mathbf{a}, \mathbf{b}}$ is the set of all possible differentiable paths in Ψ between the points $\mathbf{a}$ and $\mathbf{b}$ and $\Gamma(s) : \mathbb{R} \rightarrow \mathbb{R}^2$ indicates one such path, parametrized by its arclength $s \in [0, l(\Gamma)]$. The spatial derivative $\Gamma'(s) = \partial \Gamma(s) / \partial s$ is the unit vector tangent to the direction of the path.

This question is about the parameter γ .

Consider Figure 4. (a) shows the input image I , and (b) shows the binary mask M , with the "object" region Ω (the dark region). Determine for (c) and (d) if they are the distance map $D_l(\mathbf{x}; M, \nabla I)$ with $\gamma = 0$ or $\gamma > 0$ (darker means shorter distance). Explain your answer.

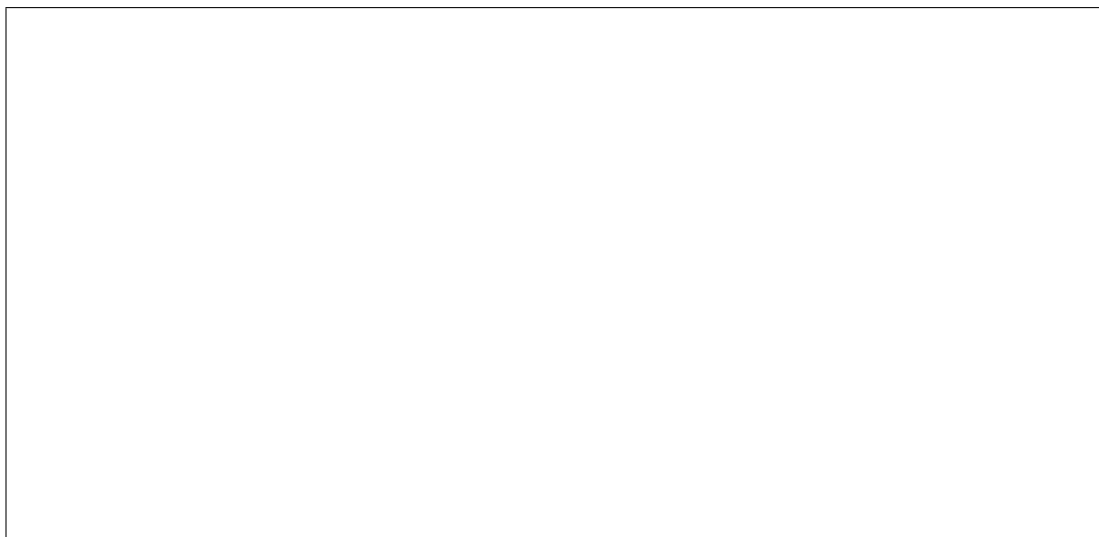
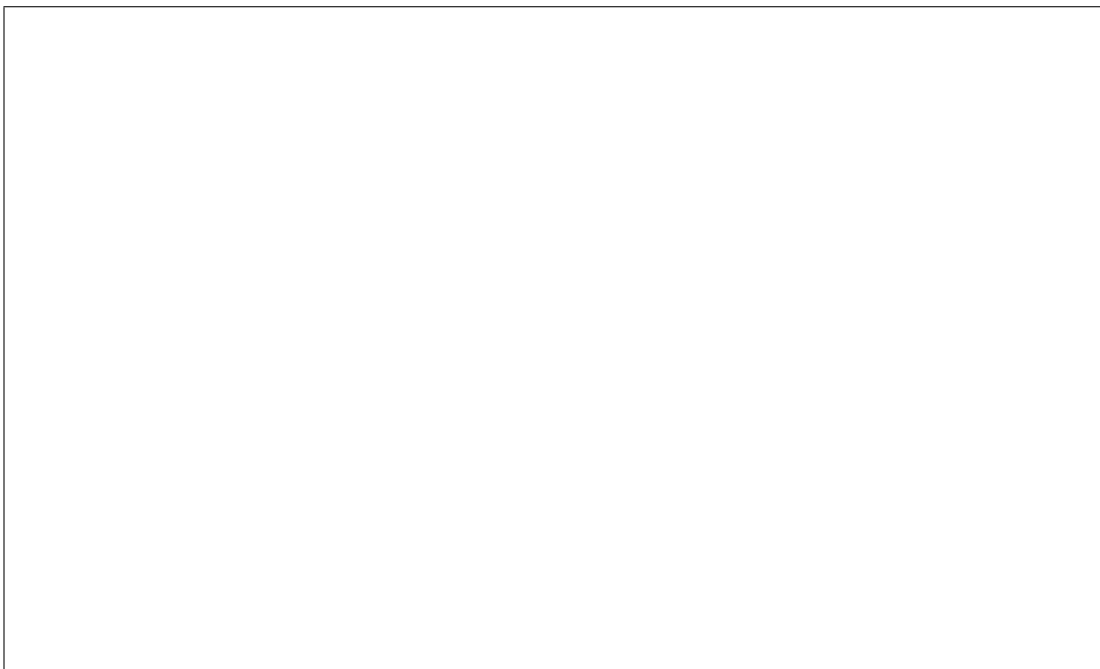




Figure 5: (Left) Two 3D point clouds with user scribbles superimposed. (Right) Segmentation results.

7. (13 points) SEGMENTATION

Consider Figure 5. The left part shows two 3D point clouds (a plane and a motorcycle) with user scribbles superimposed (Diamond shaped points with different colors). A 3D point cloud is simply an unordered list of 3D points. Suggest an algorithm that takes a 3D point cloud and user scribbles and produces the segmentation results seen on the right side of the figure.



8. (12 points) BIDIRECTIONAL SIMILARITY

Given point sets $P = \{p_1, \dots, p_n\}$ and $Q = \{q_1, \dots, q_m\}$ in some Euclidean space, define the "distance" $D(P, Q)$:

$$D(P, Q) = \max_{p \in P} d(p, Q), \text{ with} \quad (3)$$

$$d(p, Q) = \min_{q \in Q} d(p, q) \quad (4)$$

and $d(p, q)$ is the usual Euclidean distance between points p and q .

Is D symmetric? That is, show that $D(P, Q) = D(Q, P)$ or give a counter-example.

EMERGENCY BOXES

A large, empty rectangular box with a thin black border, intended for providing emergency contact information.A second large, empty rectangular box with a thin black border, identical to the one above, for additional emergency information.